Swapnil Chowdhury

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EDUCATION

Chittagong University of Engineering and Technology **July 2025**
B.Sc. in Computer Science

- GPA: 3.72 out of 4.00

PROFESSIONAL EXPERIENCE

East Delta University **Jan 2026 - Present**
Lecturer

- Conducting theory classes and lab courses for undergraduate students.

Gigatech Limited **Jan 2025**
Internship

- Gained hands-on experience in Machine Learning tasks including data preprocessing, model training, and evaluation. Explored the fundamentals and applications of Large Language Models (LLMs) in real-world scenarios.

PUBLICATIONS

1. S. Chowdhury, K. Chowdhury, R. Mukta, and S. C. Tista, "A Federated Approach to Verifier Verification in Self Sovereign Identity", in *Distributed Ledger Technology (SDLT 2025)*, vol. 2892, Springer, Singapore, 2026, pp. 16–28.

2. J. Naima, T. Chakraborty, S. Chowdhury, M. T. Khatun, M. Tripura, F. A. Tofa, "Getting the Joke: Stacking Diverse Transformers for Humor Intensity Regression in Edited News Headlines", in *MIET 2026, Springer Lecture Notes in Networks and Systems (LNNS)* (accepted, to appear).

3. I. Samiha and S. Chowdhury,"Multi-Blockchain Backup & Restore System with Zero-Knowledge Proofs (ZKPs)" , in *Proceedings of the 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM 2025)*, pp. 1–6.

4. I. Samiha, S. Chowdhury, N. Mobassara and M. I. Uddin Forkan,"A Hybrid Deep Learning Approach for Dialect Specific Sentiment Analysis of Bangla Regional Linguistics", in *Proceedings of the 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM 2025)*, pp. 1–6.

RESEARCH EXPERIENCE

Multi-Blockchain Backup Restore System with Zero-Knowledge Proofs (ZKPs)

- Designed a decentralized backup system using Ethereum and Hyperledger with IPFS for encrypted data storage.
- Used ZKPs for data integrity and Shamir's Secret Sharing for multi-party restoration.

A Hybrid Deep Learning Approach for Dialect-Specific Sentiment Analysis of Bangla Linguistics

- Focused on sentiment analysis across three Bangla dialects—Chittagong, Noakhali, and Sylhet.
- Achieved 94.69 % accuracy using a hybrid LSTM-CNN model; F1-score of 0.95 for dialect detection using Naive Bayes.

- Applied traditional ML and BERT-based deep learning for robust sentiment recognition.

A Federated Approach to Verifier Verification in Self Sovereign Identity
Undergraduate Thesis

- Proposed a federated model to allow users to verify the trustworthiness of verifiers without compromising privacy.
- Implemented a privacy-preserving credential verification system integrating blockchain (Hyperledger Fabric), Zero-Knowledge Proofs.

Supervisor: Dr. Rahma Bintey Mufiz Mukta

TECHNICAL SKILLS

Programming Languages: *C, C++, Java, Python, JavaScript*

Frameworks & Libraries: *Node.js, React.js, Tailwind CSS, TensorFlow, PyTorch, Matplotlib, Pandas, NumPy*

Machine Learning & Deep Learning: *Supervised/Unsupervised ML algorithms, CNNs, RNNs, Transformers*

Data Science & Analysis: Analysis and visualization using *Jupyter Notebooks*

Databases: *MySQL, MongoDB*

Systems & Platforms: *Linux, Hyperledger Fabric, IPFS*

PROJECTS

- **Crop Recommendation System**
  - Developed a system using Flask, Python, and machine learning that helps farmers determine the best crops to grow based on soil nutrients, temperature, humidity, pH levels, and rainfall.
- **Home-Harmony Autism**
  - Built an innovative homeschooling platform for children with autism, featuring math games, color identification, and animal/number matching games to enhance cognitive skills.
- **Blockchain-based Project**
  - Developed a blockchain-based decentralized environment using ReactJS and Tailwind CSS, ensuring secure storage of user information and transactions in an immutable ledger.
  - Participated in the Blockchain Olympiad and reached the final round.
- **Image Classification using CNN**
  - Worked on image classification using Convolutional Neural Networks (CNNs), applying deep learning techniques to accurately classify images.
- **Finding Tutor for Students**
  - Developed a web application using Vanilla JavaScript to connect students with suitable tutors based on subject preferences and location.

ACHIEVEMENTS

Blockchain Olympiad Bangladesh 2024
Finalist (Ranked 14th Nationally)